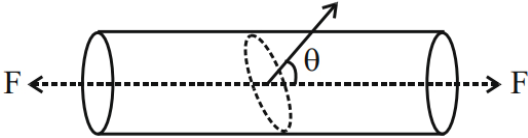
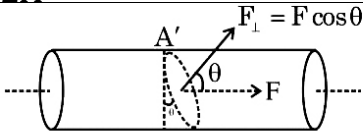
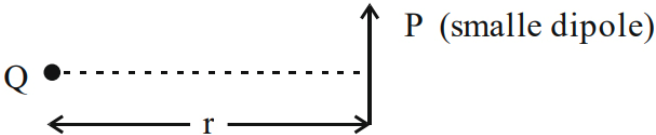
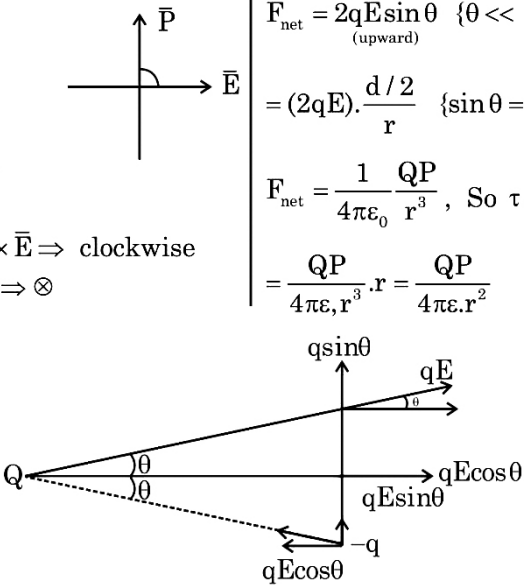
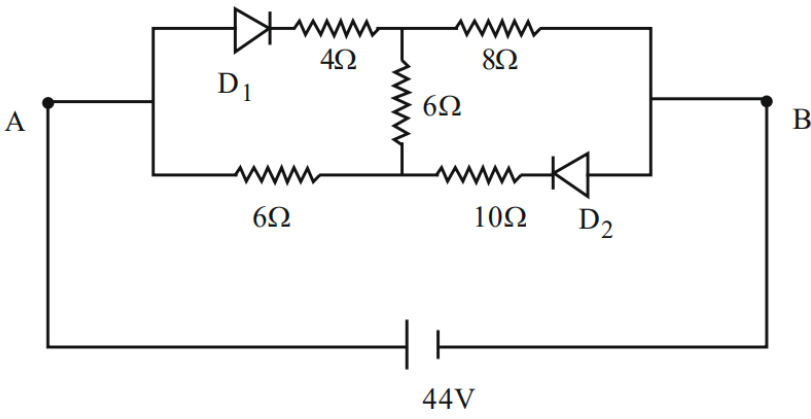
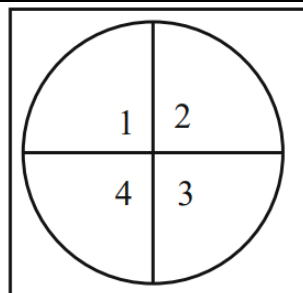
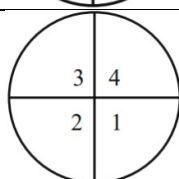
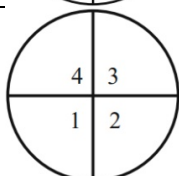
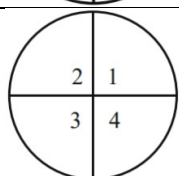
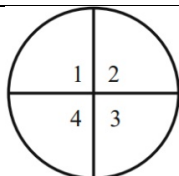


1.	A carbon dioxide laser emits a sinusoidal electromagnetic wave that travel in vacuum in the negative X-direction. The wavelength is $10.6 \mu\text{m}$ and the $\vec{E}$ field is parallel to z-axis with $E_{\text{max}} = 1.5 \text{ MV/m}$. Write vector equation of $\vec{B}$ as function of time and position.	A
	$\vec{B}(x, t) = \hat{j}(5.0 \times 10^{-3} \text{T}) \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]$	
	$\vec{B}(x, t) = -\hat{j}(5.0 \times 10^{-3}) \text{T} \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]$	
	$\vec{B}(x, t) = \hat{K}(1.5 \times 10^6 \text{V/m}) \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]$	
	$\vec{B}(x, t) = \hat{K}(1.5 \times 10^6 \text{ V/m}) \cos[(5.93 \times 10^5 \text{ rad/m})x + (1.78 \times 10^{14} \text{ rad/s})t]$	
Sol.	Wave is propagating in -ve x-direction. $\therefore \hat{E} \times \hat{B} = \hat{V}$ $\hat{K} \times \hat{B} = -\hat{i}$ So $\hat{B} = \hat{j}$ $B_0 = \frac{E_0}{C} = \frac{1.5 \times 10^6}{3 \times 10^8} = 5 \times 10^{-3} \text{T}$ $B = (B_0 \hat{j}) \sin[\omega t + kx + \phi]$ Or $\vec{B} = (B_0 \hat{j}) \cos[\omega t + kx + \phi]$ $\vec{B} = (5 \times 10^{-3} \hat{j}) \cos(\omega t + kx)$	
2.	A bar of cross section A is subjected to equal and opposite tensile forces at its ends. Consider a plane section of the bar whose normal makes an angle θ with the axis of the bar. What is the tensile stress on the plane? 	A
	$\frac{F}{A} \cos^2 \theta$	
	$\frac{F}{2A} \cos^2 \theta$	
	$\frac{F}{A} \sin 2\theta$	
	$\frac{F}{2A} \sin 2\theta$	
Sol.	 $A' \cos \theta = A$ $A' = A / \cos \theta$ Tensile stress $= \frac{F_{\perp}}{A'} = \frac{F \cos \theta}{A / \cos \theta} = \frac{F}{A} \cos^2 \theta$	
3.	For the situation shown in the figure (assume $r \gg$ length of dipole) select the correct statement.	D

		
	Force acting on dipole is zero.	
	Force acting on dipole is $\frac{PQ}{2\pi\epsilon_0 r^3}$ and is acting upwards	
	Force acting on dipole is $\frac{PQ}{2\pi\epsilon_0 r^3}$ and is acting downwards.	
	Torque acting on the dipole is $\frac{PQ}{4\pi\epsilon_0 r^2}$ Sense of rotation is clockwise	
Sol.	$\vec{\tau} = \vec{P} \times \vec{E}$ Here $\vec{P} \perp \vec{E}$ $\tau = PE \sin 90^\circ = PE$ $\tau = P \cdot \frac{KQ}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{PQ}{r^2}$ Sense of rotation $\Rightarrow \vec{P} \times \vec{E} \Rightarrow$ clockwise Direction of torque $\Rightarrow \otimes$ Force on dipole  $F_{\text{net}} = 2qE \sin \theta$ {$\theta \ll$ (upward)} $= (2qE) \cdot \frac{d/2}{r}$ {$\sin \theta = \theta$} $= \frac{Eqd}{r} = \frac{KQ}{r^2} \cdot \frac{qd}{r}$ $F_{\text{net}} = \frac{1}{4\pi\epsilon_0} \frac{QP}{r^3}$, So $\tau = Fr_1$ $= \frac{QP}{4\pi\epsilon_0 r^3} \cdot r = \frac{QP}{4\pi\epsilon_0 r^2}$	
4.	In The given circuit what is the current flow through resistance 4Ω? 	A
	3A	
	1A	
	2A	
	0A	
Sol.	D_2 is reversed bias therefore no current from 10Ω resistance.	
5.	A convex lens is used to get a real image of the object shown in the following figure.	C



Then the real inverted image is



Sol. In convex lens real image is inverted.

6. A metal rod moves at a constant velocity in a direction perpendicular to its length. A constant uniform magnetic field exists in space in a direction perpendicular to the rod as its velocity. Select the correct statement from the following:

B

The entire rod is at the same electric potential

There is an electric field in the rod

The electric potential is highest at the centre of the rod and decrease towards its ends

The electric potential is lowest at the centre of the rod and increases towards its ends.

Sol. + + High potential

$\vec{E}$ {from H.P. to L.P.}

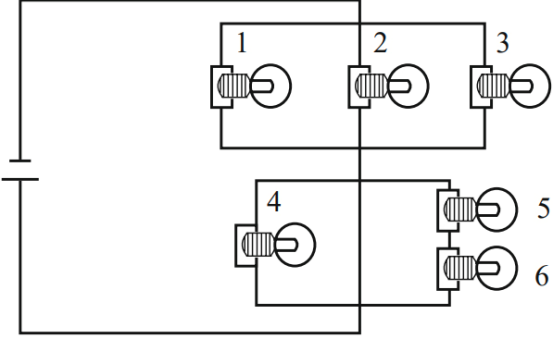
$\vec{v}$

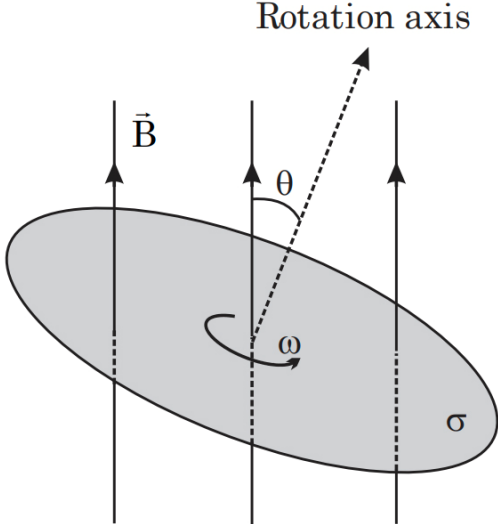
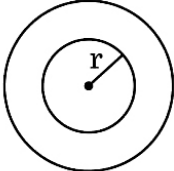
- - at low potential

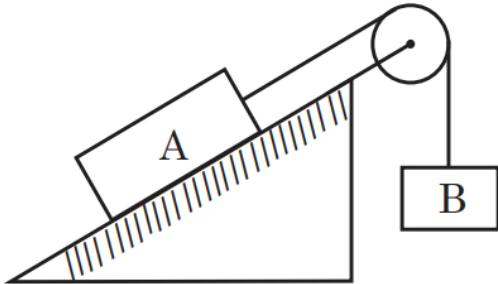
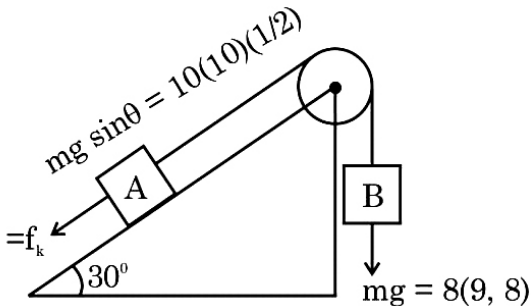
$$\text{emf induced} = \vec{\ell} \cdot [\vec{v} \times \vec{B}]$$

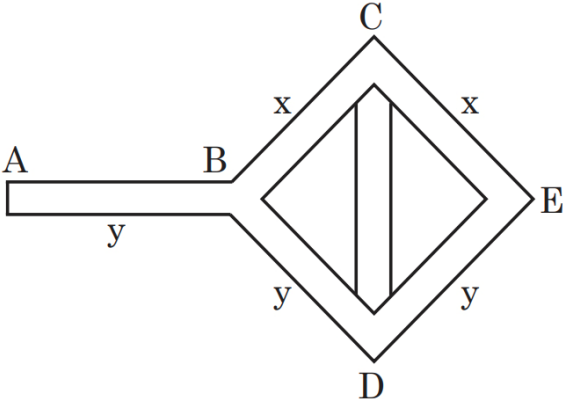
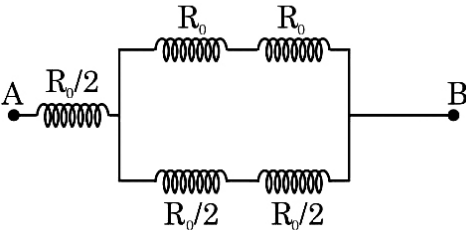
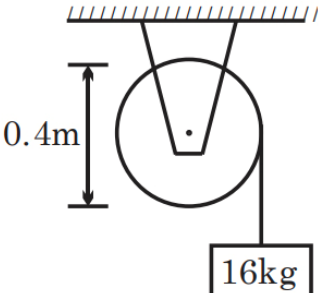
7. A dam for water reservoir is built thicker at the bottom than at the top bottom than at the top because:

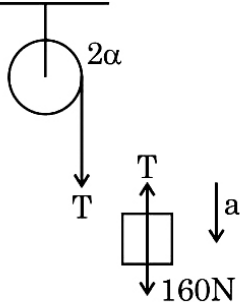
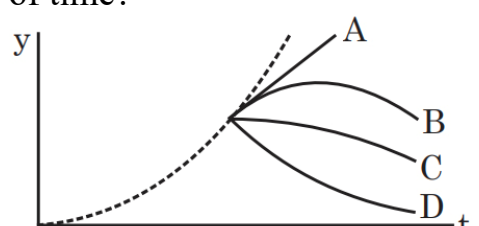
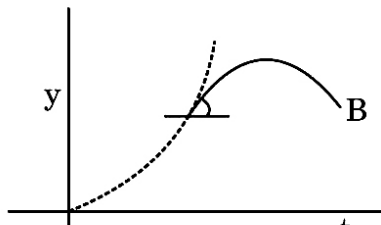
A

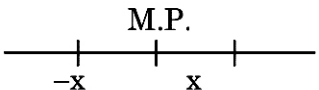
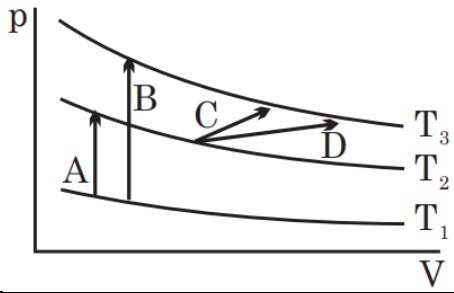
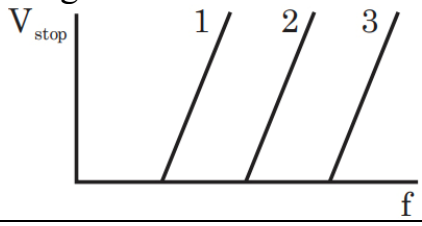
	Pressure of water is very large at the bottom due to its large depth	
	Water is likely to have more density at the bottom due to its large depth	
	Quantity of water at the bottom is large	
	None of the above	
Sol.	$P = P_0 + \rho gh$ as, $h \uparrow$, $P \uparrow$, so we require more strength at bottom.	
8.	An experiment measures quantities x, y, z and then t is calculated from the data as $t = \frac{xy^2}{z^3}$. If percentage errors in x,y and z are respectively 1%, 3%, 2%, then percentage error in t is:	D
	10%	
	4%	
	7%	
	13%	
Sol.	$t = \frac{xy^2}{z^3}$ $\frac{\Delta t}{t} = \left \frac{\Delta x}{x} \right + 2 \left \frac{\Delta y}{y} \right + 3 \left \frac{\Delta z}{z} \right = 1 + 2(3) + 3(2)$ $= 1 + 6 + 6 = 13\%$	
9.	Six identical light bulbs are connected to a battery to form the circuit shown. Which light bulb (s) glow the brightest? 	A
	1, 2 and 3	
	5 and 6	
	1, 2, 3 and 4	
	4 only	
Sol.	No current in bulb 4, 5 and 6.	
10.	During repair of the Hubble Space Telescope, an astronaut replaces a damaged solar panel during a spacewalk. Pushing the detached panel away into space, she is propelled in the opposite direction. The astronaut's mass is 60 kg and the panel's mass is 80 kg. Both the astronaut and the panel initially are at rest relative to the telescope. The astronaut then gives the panel a shove. After the shove it is moving at 0.30 m/s relative to the telescope. What is her subsequent velocity relative to the telescope? (During this operation the astronaut is tethered to the ship; for our calculations assume that the tether remains slack.)	B
	0.3 m/sec	
	0.4 m/sec	
	0.1 m/sec	
	0.7 m/sec	

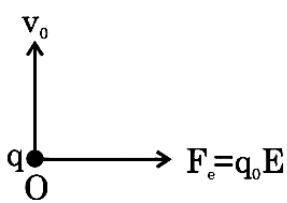
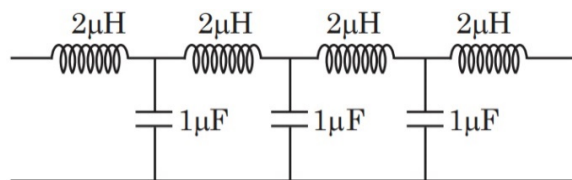
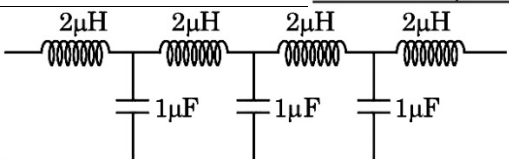
Sol.	Apply colm $\{\vec{F}_{\text{net}} = \text{zero}\}$ $m_1 v_1 + m_2 v_2 = 0$ $0.3(80) + 60(v_2) = 0$ $v_2 = \frac{-24}{60} = -0.4 \text{ m/s}$	
11.	A uniform disk of mass m, radius r, and surface charge density σ rotates about its centre with angular velocity ω in figure. A uniform magnetic field of magnitude $\vec{B}$ threads the disk, making an angle θ with respect to the rotation axis of the disk. Calculate the net torque acting on the disk.  Rotation axis	A
	$\frac{1}{4}\pi\sigma r^4 \omega B \sin \theta$	
	$\frac{3}{4}\pi\sigma r^4 \omega B \sin \theta$	
	$\frac{1}{4}\pi\sigma r^4 \omega B \cos \theta$	
	$\frac{3}{4}\pi\sigma r^4 \omega B \sin \theta$	
Sol.	 $m = iA$ $i = \frac{dQ}{dt} = \frac{\sigma 2\pi r dr}{T}$ $m = \frac{\sigma(2\pi r) dr}{T} \cdot \pi r^2$ $= \int_0^R \sigma \omega (\pi) r^3 \cdot dr = \frac{\sigma \omega \pi}{4} R^4$ $ \tau = \vec{m} \times \vec{B} = mB \sin \theta$ $= \frac{1}{4} \sigma \omega \pi R^4 B \sin \theta$	
12.	Consider the statement: “Earth moves in a stable orbit around the Sun and is therefore in equilibrium”. The statement is	D
	False, because no moving body can be in equilibrium	

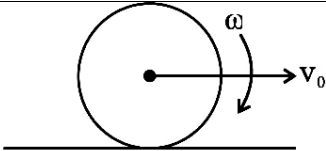
	True, because Earth does not fall into or fly away from the Sun.	
	False, because earth is rotating on its axis and no rotating body can be in equilibrium.	
	False, because earth has a considerable acceleration.	
Ans.	False, because earth has a considerable acceleration.	
13.	Block A, with a mass of 10 kg, rests on a 30° incline. The coefficient of kinetic friction is 0.20. The attached string is parallel to the incline and passes over a massless, frictionless pulley at the top. Block B, with a mass of 8.0 kg, is attached to the dangling end of the string. The acceleration of B is (approximately) 	B
	0.70 m/s ² , up	
	0.70 m/s ² , down	
	2.6 m/s ² , up	
	2.6 m/s ² , down	
Sol.	Net pulling force along string  $= 0.2(10)(9.8) \frac{\sqrt{3}}{2}$ Net pulling force along string $= 8(9.8) - 10(9.8) \cdot \frac{1}{2} - 0.2(10)(9.8) \frac{\sqrt{3}}{2}$ $a = \frac{F_{\text{net}}}{M_{\text{total}}} = \frac{F_{\text{net}}}{18} = \frac{12.42}{18}$ $\approx 0.69 \text{ m/s}^2 \text{ (downward)}$	
14.	Two rods of material x and three rods of material y are connected as shown in figure. All the rods are of identical length and cross-section. If the end A is maintained at 60°C and the junction E at 10°C , Find effective thermal Resistance. Given length of each rod = ℓ , area of cross-section = A, conductivity of x = K and conductivity of y = 2K.	B

			
	$\frac{4\ell}{3KA}$		
	$\frac{7\ell}{6KA}$		
	$\frac{4KA}{3\ell}$		
	$\frac{7KA}{3\ell}$		
Sol.	It is equivalent to balanced wheat stone bridge $R = \frac{L}{KA}$ 		$R_x = \frac{L}{KA} = R_0$ $R_y = \frac{L}{2KA} = \frac{R_0}{2}$ $R_a = \frac{R_0}{2} + \frac{2R; R_0}{3R_0}$ $= R_0 \left[\frac{1}{2} + \frac{2}{3} \right] = \frac{7R_0}{6} = \frac{7}{6} \cdot \frac{L}{KA}$
15.	A 16 kg block is attached to a cord that is wrapped around the rim of the flywheel of diameter 0.40 m and hangs vertically, as shown. The rotational inertia of the flywheel is $0.50 \text{ kg} \times \text{m}^2$. When the block is released and the cord unwinds, the acceleration of the block is 		B
	0.15 g		
	0.56 g		
	0.84 g		
	g		

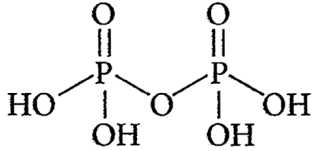
Sol.	 $160 - T = 16a$(1) $T(0.2) = 0.5(\alpha)$ {$a = \alpha R$}	$T(0.2) = 0.5 \left(\frac{a}{0.2} \right)$ $T = 2.5 \left[\frac{a}{0.2} \right] = 12.5a$(2) From eqn. (1) and (2) $160 = T + 16a = 28.5 a$ $a = \frac{160}{28.5} \text{ m/s}^2$	
16.	An elevator is moving upwards with constant acceleration. The dashed curve shows the position y of the ceiling of the elevator as a function of the time t. A bolt loose and drops from the ceiling. Which curve best represents the position of the bolt as a function of time? 	B	
	A		
	B		
	C		
	D		
Sol.	as bolt looses from ceiling. It still have vel. equal to vel. of lift at that moment. But after this, Vel. of bolt decreases bcoz of gravity and after some time it moves downward. 		
17.	A particle is in simple harmonic motion along the x axis. The amplitude of the motion is x_m. When it is at $x = x_1$ its kinetic energy is $K = 5\text{J}$ and its potential energy (measured with $U = 0$ at $x = 0$) is $U = 3\text{J}$. When it is at $x = -x_1$, the kinetic and potential energies are:	A	
	$K = 5\text{ J}$ and $U = 3\text{J}$		
	$K = 5\text{ J}$ and $U = -3\text{J}$		
	$K = 8\text{ J}$ and $U = 0$		
	$K = 0$ and $U = 8\text{J}$		

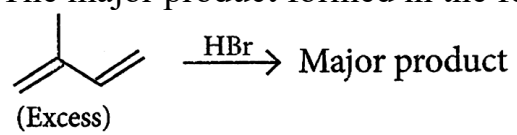
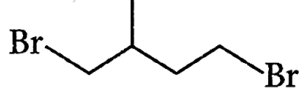
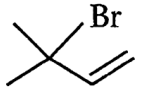
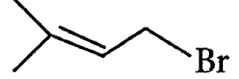
Sol.	$v = \omega\sqrt{A^2 - X^2}$ Speed is same at same separation from M.P.,  So KE at x = KE at -x So PE at x = PE at -x KE = 5J u = 3J	
18.	The diagram shows four thermodynamic processes carried out on an ideal gas. For which of these processes is the change in the internal energy of the gas the greatest? 	B
	A	
	B	
	C	
	D	
Sol.	Here $T_3 > T_2 > T_1$ $\Delta U = nC_v\Delta T$ Here ΔT is maximum in process B.	
19.	The diagram shows shown the graph of the stopping potential as a function of the frequency of the incident light for photoelectric experiments performed on three different materials. Rank the materials according to the values of their work functions, from least to greatest. 	A
	1, 2, 3	
	3, 2, 1	
	2, 3, 1	
	2, 1, 3	
Sol.	Maximum frequency is required for material 3, so $\phi_3 > \phi_2 > \phi_1$.	
20.	A charged particle q_0 of mass m_0 is projected along the y-axis at $t = 0$ from origin with velocity v_0 . There is a uniform electric field E_0 along x-axis. If time when de-Broglie wavelength becomes half of initial value is $n \frac{m_0 v_0}{q_0 E_0}$, then find n.	A
	$\sqrt{3}$	

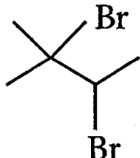
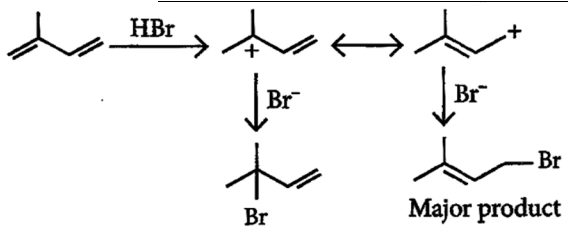
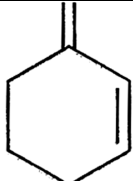
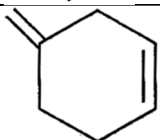
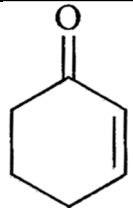
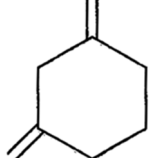
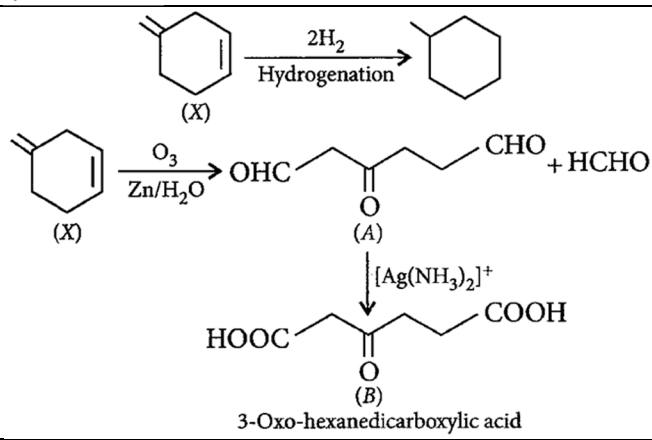
	$\sqrt{5}$	
	$\sqrt{2}$	
	$\sqrt{7}$	
Sol.	$\lambda = \frac{h}{p}$  $p_i = mv_0$ To half the wavelength p has to be doubled $v_f = 2v_0 = \sqrt{v_x^2 + v_y^2}$ $4v_0^2 = \left[\frac{q_0 E}{m} \cdot t \right]^2 + v_0^2$ $3v_0^2 = \frac{q_0^2 E^2}{m^2} \cdot t^2$ $t = \frac{mv_0}{q_0 E_0} \sqrt{3}$ $n = \sqrt{3} = 1.73$	
21.	Figure the given infinite ladder, what is the frequency at which the circuit behaves as a purely inductive circuit? If ω is $\sqrt{x} \times 10^6$ rad/sec, then find value of x. 	2
Sol.	 $Z = X_L + \frac{ZX_C}{Z + X_C}$ $Z(Z + X_C) = X_L(Z + X_C) + ZX_C$ $Z^2 + ZX_C = ZX_L + X_L X_C + ZX_C$ $Z^2 - ZX_L - X_L X_C = 0$ Solving, $Z = \omega LC \pm \sqrt{(\omega LC)^2 - 4LC}$ $2C$ For pure inductive nature, no X_C should exist So, $\omega^2 L^2 C^2 - 4LC = 0$ or $\omega = \frac{2}{\sqrt{LC}}$ $\omega = \frac{2}{(2 \times 10^{-16} \times 1 \times 10^{-6})^{1/2}} = \sqrt{2} \times 10^6 \text{ rad/s}$	
22.	Two long strings A and B each having linear mass density $1.2 \times 10^{-2} \text{ kg/m}$ are stretched by different tensions 4.8 N and 7.5 N respectively and are kept parallel to each other with their left ends at $x = 0$. Wave pulses are produced on left ends at $t = 0$ on A and $t = 20 \text{ ms}$ on B. If $n \times 10^{-2} \text{ s}$ is the time required to meet, then find n .	10
Sol.	$v_A = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{4.8}{1.2 \times 10^{-2}}} = 20 \text{ m/s}$ $v_B = \sqrt{\frac{7.5}{1.2 \times 10^{-2}}} = \sqrt{\frac{2500}{4}} = \frac{50}{2} = 25 \text{ m/s}$ $\text{Separation} = v_{\text{rel}} (\Delta t)$ $20(20) \times 10^{-3} = 5(\Delta t)$ $(\Delta t) = 80 \times 10^{-3} = 8 \times 10^{-2}$ $t_{\text{final}} = 20 \text{ ms} + 80 \text{ ms}$ $= 100 \text{ ms}$ $= 10 \times 10^{-2} \text{ sec}$	
23.	A prism having refractive index $\sqrt{2}$ has angle of minimum deviation equal to	1

	the angle of the prism, if the angle of the prism is α then fill the value of $\sin \alpha$.	
Sol.	$\mu = \sqrt{2}$ for min. deviation $i = e$ $r_1 = r_2 = A/2$ So $\delta = i + e - A$ $\delta_{\min} = 2i - A$ $\alpha = 2i - \alpha$ $i = \alpha$ Now apply Snell's law $(1)\sin i = \sqrt{2}\sin(r)$	$1\sin(\alpha) = \sqrt{2}\sin\left(\frac{\alpha}{2}\right)$ $\sin \alpha = \sqrt{2}\sin\left(\frac{\alpha}{2}\right)$ $2\sin\left(\frac{\alpha}{2}\right) \cdot \cos\left(\frac{\alpha}{2}\right) = \sqrt{2}\sin\left(\frac{\alpha}{2}\right)$ $\cos\left(\frac{\alpha}{2}\right) = \frac{1}{\sqrt{2}}$ $\frac{\alpha}{2} = 45^\circ$ $\alpha = 90^\circ$ So $\sin(\alpha) = \sin(90^\circ) = 1$
24.	A uniform disc of mass m , radius R is rolling with angular velocity ω on a horizontal plane. If angular momentum of body about a point on the surface and along the direction of motion of disc is $\frac{n}{2}mR^2\omega^2$, find n .	3
Sol.	 $\omega = \frac{v}{R}$ $L = mv_c R \otimes + I_{cm} \omega \otimes$ $= m(\omega R)(R) + \frac{mR^2}{2} \cdot \omega$ $L = \frac{3}{2}m\omega R^2$	
25.	A 1024 Hz tuning fork is used to obtain a series of resonance levels in a gas column of variable length, with one end closed and the other open. The length of the column changes by 20 cm from resonance to resonance. From this data, the speed of sound in this gas is α then fill the value of 10α .	4096
Sol.	Given $\frac{\lambda}{2} = 20\text{cm}$ $\lambda = 40\text{cm} = 0.4\text{m}$ $v = f\lambda = 1024(0.4)$ $v = 409.6 \text{ m/s}$ So $\frac{v}{100} = 4.096 \approx 4.10$	
26.	The amount of sugar ($C_{12}H_{22}O_{11}$) required to prepare 2 L of its 0.1 M aqueous solution is :	C
	136.8 g	
	34.2 g	

	68.4 g																					
	17.1 g																					
Sol.	We know, $M = \frac{\text{wt.}}{\text{Mol. wt.}} \times \frac{1}{V(\text{L})}$ $0.1 = \frac{m}{342} \times \frac{1}{2} \Rightarrow m = 342 \times 2 \times 0.1 = 68.4 \text{ g}$																					
27.	If the radius of 3 rd Bohr's orbit of hydrogen atom is r_3 and the radius of 4 th Bohr's orbit is r_4 , then	B																				
	$r_4 = \frac{9}{16} r_3$																					
	$r_4 = \frac{16}{9} r_3$																					
	$r_4 = \frac{3}{4} r_3$																					
	$r_4 = \frac{4}{3} r_3$																					
Sol.	We know, $r_n = \frac{0.529 \times n^2}{Z} \text{ \AA}$ [n = number of shell, Z = atomic number] $Z = 1$ (for H-atom) Radius of 3 rd Bohr's orbit of hydrogen atom ($n = 3$) $r_3 = 0.529 \times 9 \text{ \AA}$ (i) Radius of 4 th Bohr's orbit of hydrogen atom ($n = 4$) $r_4 = 0.529 \times 16 \text{ \AA}$... (ii) From equation (i) and (ii), we get $\frac{r_3}{r_4} = \frac{0.529 \times 9 \text{ \AA}}{0.529 \times 16 \text{ \AA}} = \frac{9}{16} \Rightarrow r_4 = \frac{16}{9} r_3$																					
28.	Match List-I and List-II. <table border="1"><thead><tr><th colspan="2">List I</th><th colspan="2">List II</th></tr></thead><tbody><tr><td>A.</td><td>Osmosis</td><td>I.</td><td>Solvent molecules pass through semi permeable membrane towards solvent side.</td></tr><tr><td>B.</td><td>Reverse osmosis</td><td>II.</td><td>Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes .</td></tr><tr><td>C.</td><td>Electroosmosis</td><td>III.</td><td>Solvent molecules pass through semi permeable membrane towards solution side.</td></tr><tr><td>D.</td><td>Electrophoresis</td><td>IV.</td><td>Dispersion medium moves in an electric field.</td></tr></tbody></table> Choose the correct answer from the options given below :	List I		List II		A.	Osmosis	I.	Solvent molecules pass through semi permeable membrane towards solvent side.	B.	Reverse osmosis	II.	Movement of charged colloidal particles under the influence of applied electric potential towards oppositely charged electrodes .	C.	Electroosmosis	III.	Solvent molecules pass through semi permeable membrane towards solution side.	D.	Electrophoresis	IV.	Dispersion medium moves in an electric field.	D
List I		List II																				
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D.	Electrophoresis	IV.	Dispersion medium moves in an electric field.																			
	A-I, B-III, C-IV, D-II																					
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	A-I, B-III, C-II, D-IV																					
	A-III, B-I, C-II, D-IV																					

Sol.	A-III, B-I, C-II, D-IV	
29.	Which one of the following statements is not true?	C
	The conjugate base of H_2PO_4^- is HPO_4^{2-}	
	$\text{pH} + \text{pOH} = 14$ for all aqueous solutions	
	The pH of 1×10^{-8} M HCl is 8	
	96,500 coulombs of electricity when passed through a CuSO_4 solution deposits 1 gram equivalent of copper at the cathode.	
Sol.	pH of an acid cannot exceed 7. Here we should also consider $[\text{H}^+]$ that comes from H_2O. Now $[\text{H}^+] = [\text{H}^+]_{\text{from HCl}} + [\text{H}^+]_{\text{from H}_2\text{O}}$ $= 10^{-8} + 10^{-7} = 10^{-8} + 10 \times 10^{-8} = 11 \times 10^{-8}$ $\therefore \text{pH} = -\log(11 \times 10^{-8}) = 6.9587$	
30.	Aluminium oxide may be electrolysed at 1000°C to furnish aluminium metal (At. Mass = 27 amu; 1 Faraday = 96,500 Coulombs). The cathode reaction is $\text{Al}^{3+} + 3\text{e}^- \rightarrow \text{Al}^0$ To prepare 5.12 kg of aluminium metal by this method would require	A
	5.49×10^7 C of electricity	
	1.83×10^7 C of electricity	
	5.49×10^4 C of electricity	
	5.49×10^{10} C of electricity	
Sol.	From Faraday's 1st law, $W = Z \times Q$ [W = weight, Z = electrochemical equivalent, Q = quantity of electricity] Now $E = Z \times F$ [E = equivalent weight, F = Faraday] $w = \frac{E}{F} \times Q \Rightarrow Q = \frac{w \times F}{E} = \frac{w \times F}{\frac{A}{n}}$ [A = Atomic weight, n = valency of ion] or $Q = \frac{n \times w \times F}{A} = \frac{3 \times 5.12 \times 10^3 \times 96500}{27} = 5.49 \times 10^7 \text{ C}$	
31.	In a molecule of pyrophosphoric acid, the number of P – OH, P = O and P – O – P bonds/moiety(ies) respectively are	D
	2, 4 and 1	
	3, 3 and 3	
	4, 2 and 0	
	4, 2 and 1	
Sol.	 Pyrophosphoric acid Number of P – OH bonds = 4 Number of P = O bonds = 2 Number of P – O – P bond = 1	
32.	The colour of KMnO_4 is due to	A
	$\text{L} \rightarrow \text{M}$ charge transfer transition	
	$\sigma \rightarrow \sigma^*$ transition	

	M → L charge transfer transition	
	d – d transition	
Sol.	The deep purple colour of KMnO ₄ is not due to d – d transitions but due to charge transfer from O to Mn (i.e., L → M), which reduces the oxidation state of Mn from +7 to +6 momentarily.	
33.	On treatment of 100 mL of 0.1 M solution of CoCl ₃ .6H ₂ O with excess AgNO ₃ , 1.2 × 10 ²² ions are precipitated. The complex is	B
	[Co(H ₂ O) ₆]Cl ₃	
	[Co(H ₂ O) ₅ Cl]Cl ₂ .H ₂ O	
	[Co(H ₂ O) ₄ Cl ₂]Cl.2H ₂ O	
	[Co(H ₂ O) ₃ Cl ₃].3H ₂ O	
Sol.	Number of moles of complex $= \frac{M \times V(\text{mL})}{1000} = \frac{0.1 \times 100}{1000} = 0.01$ Moles of ions precipitated with excess of AgNO₃ $= \frac{1.2 \times 10^{22}}{6.022 \times 10^{23}} = 0.01992 \approx 0.02$ Now, number of Cl⁻ ions present in ionisation sphere $= \frac{\text{Moles of ions precipitated with excess AgNO}_3}{\text{Moles of complex}}$ $= \frac{0.02}{0.01} = 2$ Hence, the formula of complex is [Co(H₂O)₅Cl]Cl₂.H₂O	
34.	A flask contains a mixture of isohexane and 3-methylpentane. One of the liquids boils at 63°C while the other boils at 60°C. What is the best way to separate the two liquids and which one will be distilled out first.	A
	Fractional distillation, isohexane	
	Simple distillation, 3-methylpentane	
	Fractional distillation, 3-methylpentane	
	Simple distillation, isohexane	
Sol.	The difference in the b.pt. of two liquids is very small hence fractional distillation method is used. Liquids having lower b.pt. comes out first. Therefore, iso-hexane (b.pt. 60°C) comes out first.	
35.	The major product formed in the following reaction is	C
	 (Excess)	
		
		
		

		
Sol.		
36.	An unsaturated hydrocarbon X absorbs two hydrogen molecules and catalyst hydrogenation, and also gives following reaction : $X \xrightarrow[\text{Zn/H}_2\text{O}]{\text{O}_3} A \xrightarrow{[\text{Ag}(\text{NH}_3)_2]^+} B$ (3-oxohexanedicarboxylic acid) X will be	B
		
		
		
		
Sol.		
37.	Match List-I with List-II 1-Bromopropane is reacted with reagents in List I to give product in List II	B

List I (Reagent)		List II (Product)	
A.	KOH (alc.)	I.	Nitrile
B.	KCN (alc.)	II.	Ester
C.	AgNO ₂	III.	Alkene
D.	H ₃ CCOOAg	IV.	Nitroalkane

Choose the correct answer from the options given below :

A-I, B-II, C-IV, D-II

A-III, B-I, C-IV, D-II

A-I, B-II, C-III, D-IV

A-IV, B-III, C-II, D-I

Sol.

$$\begin{array}{l}
 \text{Br}-\text{CH}_2-\text{CH}_2-\text{CH}_3 \xrightarrow{\text{KOH}/(\text{alc.})} \text{CH}_2=\text{CH}-\text{CH}_3 \\
 \hspace{10em} \text{Alkene} \\
 \hspace{10em} \xrightarrow{\text{KCN}/(\text{alc.})} \text{CH}_3\text{CH}_2\text{CH}_2\text{CN} \\
 \hspace{10em} \text{Nitrile} \\
 \hspace{10em} \xrightarrow{\text{AgNO}_2} \text{CH}_3\text{CH}_2\text{CH}_2\text{NO}_2 \\
 \hspace{10em} \text{Nitroalkane} \\
 \hspace{10em} \xrightarrow{\text{CH}_3\text{COOAg}} \text{CH}_3-\text{COOCH}_2\text{CH}_2\text{CH}_3 \\
 \hspace{10em} \text{Ester}
 \end{array}$$

38. Among the compounds A and B with molecular formula C₉H₁₈O₃, A is having higher boiling point than B. The possible structures of A and B are

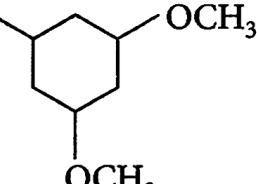
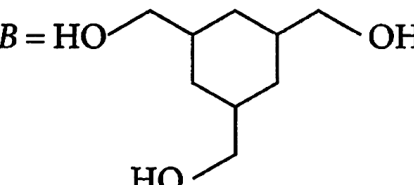
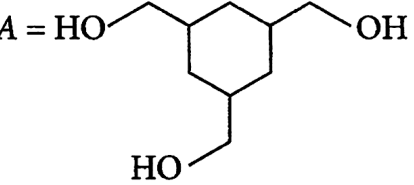
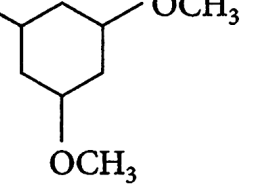
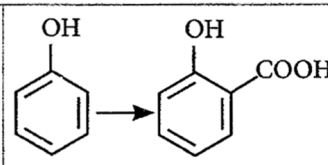
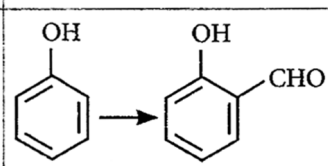
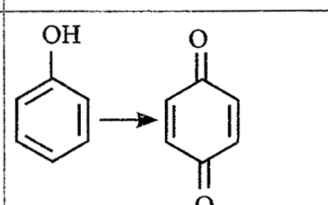
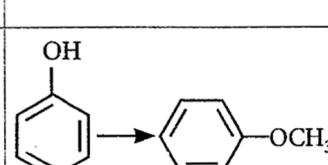
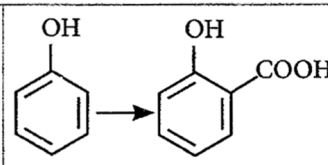
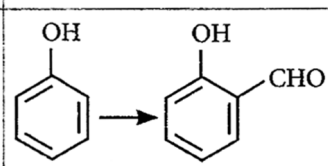
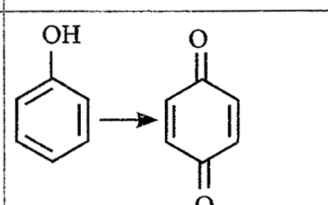
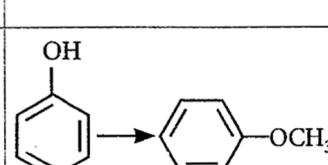
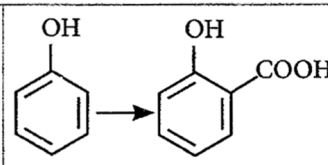
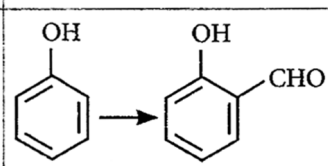
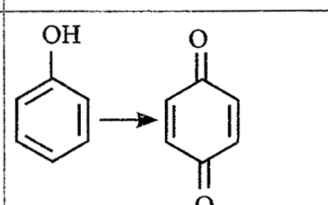
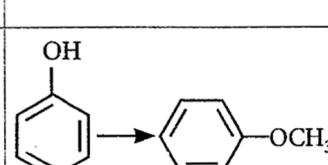
D

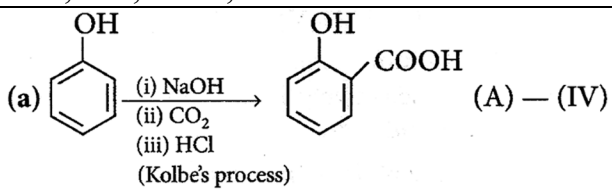
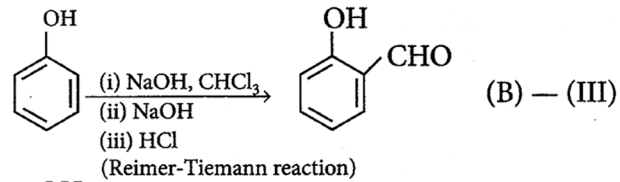
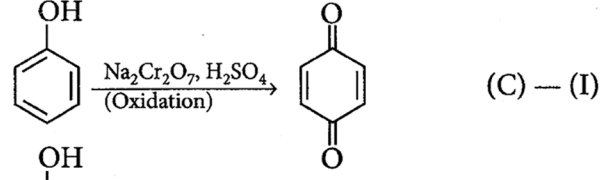
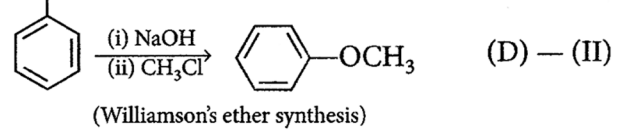
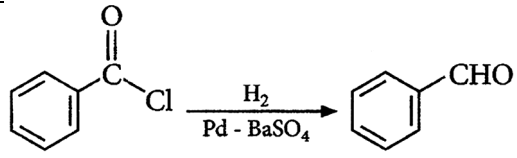
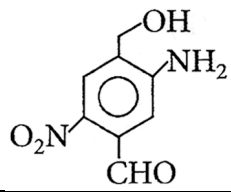
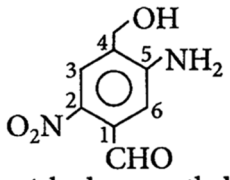
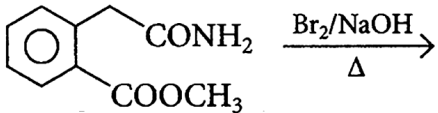
A =

B =

A =

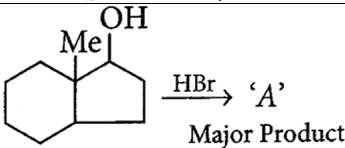
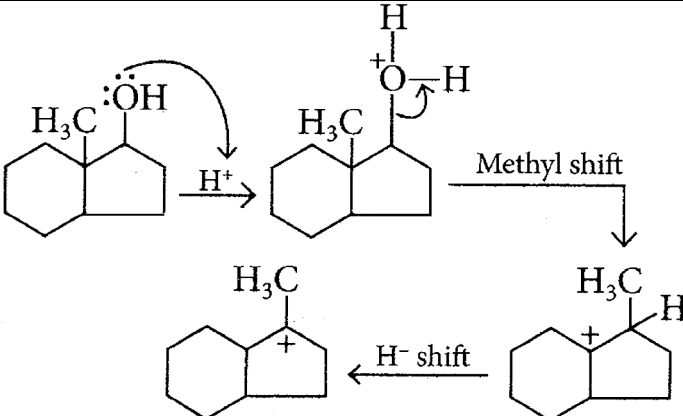
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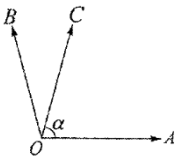
	$A = \text{H}_3\text{CO}-\text{C}_6\text{H}_3(\text{OCH}_3)_2$  $B = \text{HO}-\text{CH}_2-\text{C}_6\text{H}_3(\text{CH}_2\text{OH})_2$ 																					
	$A = \text{HO}-\text{CH}_2-\text{C}_6\text{H}_3(\text{CH}_2\text{OH})_2$  $B = \text{H}_3\text{CO}-\text{C}_6\text{H}_3(\text{OCH}_3)_2$ 																					
Sol.	In compound (A) intermolecular H-bonding is possible by virtue of which, it has higher boiling point.																					
39.	Match List-I with List-II.	A																				
<table border="1"> <thead> <tr> <th colspan="2">List-I (Reaction)</th> <th colspan="2">List-II (Reagent(s))</th> </tr> </thead> <tbody> <tr> <td>(A)</td> <td>  </td> <td>(I)</td> <td>$\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4</td> </tr> <tr> <td>(B)</td> <td>  </td> <td>(II)</td> <td>(i) NaOH (ii) CH_3Cl</td> </tr> <tr> <td>(C)</td> <td>  </td> <td>(III)</td> <td>(i) NaOH, CHCl_3 (ii) NaOH (iii) HCl</td> </tr> <tr> <td>(D)</td> <td>  </td> <td>(IV)</td> <td>(i) NaOH (ii) CO_2 (iii) HCl</td> </tr> </tbody> </table>			List-I (Reaction)		List-II (Reagent(s))		(A)		(I)	$\text{Na}_2\text{Cr}_2\text{O}_7$, H_2SO_4	(B)		(II)	(i) NaOH (ii) CH_3Cl	(C)		(III)	(i) NaOH , CHCl_3 (ii) NaOH (iii) HCl	(D)		(IV)	(i) NaOH (ii) CO_2 (iii) HCl
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Choose the correct answer from the options given below :																						
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A-IV, B-II, C-I, D-III																						

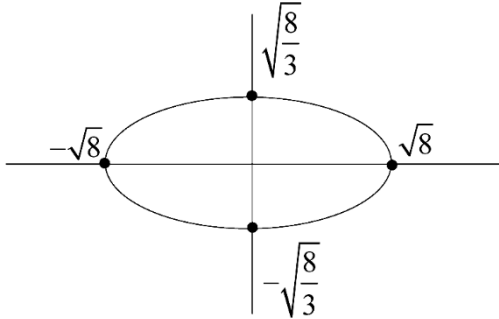
	A-II, B-I, C-III, D-IV	
Sol.	(a)    	
40.	 This reduction reaction is known as	C
	Wolff-Kishner Reduction	
	Etard Reduction	
	Rosenmund Reduction	
	Stephen Reduction	
Sol.	The given reaction is known as Rosenmund reduction.	
41.	The IUPAC name of the following compound is 	C
	2-nitro-4-hydroxymethyl-5-aminobenzaldehyde	
	3-amino-4-hydroxymethyl-5-nitrobenzaldehyde	
	5-amino-4-hydroxymethyl-2-nitrobenzaldehyde	
	4-amino-2-formyl-5-hydroxymethylnitrobenzene.	
Sol.	 5-Amino - 4-hydroxymethyl-2-nitrobenzaldehyde	
42.	The major product formed in the following reaction is 	B

Sol.	(Hoffmann bromamide degradation)																					
43.	Match List-I with List-II. <table><thead><tr><th colspan="2">List I (Reaction)</th><th colspan="2">List II (Reagents)</th></tr></thead><tbody><tr><td>A.</td><td>Hoffmann Degradation</td><td>I.</td><td>Conc. KOH, Δ</td></tr><tr><td>B.</td><td>Clemmensen Reduction</td><td>II.</td><td>CHCl₃, NaOH/H₃O⁺</td></tr><tr><td>C.</td><td>Cannizzaro Reaction</td><td>III.</td><td>Br₂, NaOH</td></tr><tr><td>D.</td><td>Reimer-Tiemann Reaction</td><td>IV.</td><td>Zn-Hg/HCl</td></tr></tbody></table> Choose the correct answer from the options given below: A-III, B-IV, C-II, D-I A-III, B-IV, C-I, D-II A-II, B-IV, C-I, D-III A-II, B-I, C-III, D-IV	List I (Reaction)		List II (Reagents)		A.	Hoffmann Degradation	I.	Conc. KOH, Δ	B.	Clemmensen Reduction	II.	CHCl ₃ , NaOH/H ₃ O ⁺	C.	Cannizzaro Reaction	III.	Br ₂ , NaOH	D.	Reimer-Tiemann Reaction	IV.	Zn-Hg/HCl	B
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Sol.	Hoffmann degradation – Br₂, NaOH Clemmensen reduction – Zn-Hg/HCl Cannizzaro reaction – conc. KOH, Δ Reimer-Tiemann reaction – CHCl₃, NaOH/H₃O⁺																					
44.	Match List-I with List-II.	B																				

	List I (Test)		List II (Functional group/Class of Compound)		
	A.	Molisch's Test	I.	Peptide	
	B.	Biuret Test	II.	Carbohydrate	
	C.	Carbylamine Test	III.	Primary amine	
	D.	Schiff's Test	IV.	Aldehyde	
Choose the correct answer from the options given below:					
	A-III, B-IV, C-II, D-I				
	A-II, B-I, C-III, D-IV				
	A-III, B-IV, C-I, D-II				
	A-I, B-II, C-III, D-IV				
Sol.	Molisch's test – Carbohydrate (A-II) Biuret test – Peptide (B-I) Carbylamine test – Primary amine (C-III) Schiff's test – Aldehyde (D-IV)				
45.	For a dimerization reaction, $2A(g) \rightarrow A_2(g)$, at 298 K, $\Delta U^\circ = -20 \text{ kJ mol}^{-1}$, $\Delta S^\circ = -30 \text{ J K}^{-1} \text{ mol}^{-1}$, then the ΔG° will be _____ J.				A
	– 13538				
	+ 13538				
	– 14738				
	+ 15338				
Sol.	$\Delta H^\circ = \Delta U^\circ + \Delta n_g RT$ As $\Delta n_g = (1 - 2) = -1$ $\therefore \Delta H = -20 \times 1000 \text{ J mol}^{-1} + ((-1) \times 8.314 \text{ JK}^{-1} \text{ mol}^{-1} \times 298 \text{ K})$ $= -22477.572 \text{ J mol}^{-1}$ $\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$ $= -22477.572 - (298 \text{ K}) \times (-30 \text{ J mol}^{-1} \text{ K}^{-1})$ $= -13537.572 \text{ J mol}^{-1} \approx -13538 \text{ J mol}^{-1}$				
46.	The total number of electron in all bonding molecular orbitals of O_2^{2-} is _____. (Round of to the Nearest Integer).				10
Sol.	$O_2^{2-} : \sigma 1s^2, \sigma^* 1s^2, \sigma 2s^2, \sigma^* 2s^2, \sigma 2p_z^2, \pi 2p_x^2 = \pi 2p_y^2, \pi^* 2p_x^2 = \pi^* 2p_y^2$ Total electrons in all bonding molecular orbitals = 10.				
47.	At 298 K, the equilibrium constant is 2×10^{15} for the reaction : $Cu(s) + 2Ag^+(aq) \rightleftharpoons Cu^{2+}(aq) + 2Ag(s)$ The equilibrium constant for the reaction $\frac{1}{2} Cu^{2+}(aq) + Ag(s) \rightleftharpoons \frac{1}{2} Cu(s) + Ag^+(aq)$ is $x \times 10^{-8}$. The value of x is _____. (Nearest Integer)				2
Sol.	$Cu_{(s)} + 2Ag_{(aq)}^+ \rightleftharpoons Cu_{(aq)}^{2+} + 2Ag_{(s)}; K_c = 2 \times 10^{15}$ $\frac{1}{2} Cu_{(aq)}^{2+} + Ag_{(s)} \rightleftharpoons \frac{1}{2} Cu_{(s)} + Ag_{(aq)}^+; K'_c = \left(\frac{1}{K_c}\right)^{1/2}$ $K'_c = \left(\frac{1}{2 \times 10^{15}}\right)^{1/2} = (5 \times 10^{-16})^{1/2} = 2.24 \times 10^{-8}$ The value of 'x' is 2.				

48.	Number of amphoteric compounds among the following is _____. (A) BeO (B) BaO (C) Be(OH) ₂ (D) Sr(OH) ₂	2
Sol.	BeO — Amphoteric oxide BaO — Basic oxide Be(OH) ₂ — Amphoteric hydroxide Sr(OH) ₂ — Basic hydroxide	
49.	 The number of hyperconjugation structure involved to stabilize carbocation formed in the above reaction is _____.	6
Sol.	 (Stabilised carbocation) Due to this structure, there are 7 hyperconjugation structure.	
50.	$\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \begin{cases} \xrightarrow{\text{C}_2\text{H}_5\text{OH}} \text{Product A} \\ \xrightarrow{\text{H}_2\text{O}} \text{Product B} \end{cases}$ The total number of hydrogen atoms in product A and product B is _____.	10
Sol.	$\text{CH}_3\text{CH}_2\text{Br} + \text{NaOH} \begin{cases} \xrightarrow{\text{C}_2\text{H}_5\text{OH}} \text{CH}_2=\text{CH}_2 \text{ (A)} \\ \xrightarrow{\text{H}_2\text{O}} \text{CH}_3\text{CH}_2\text{OH} \text{ (B)} \end{cases}$ Hence, the total number of hydrogen atoms in product A and product B is 10.	
51.	$f(x) = \begin{cases} 4x - x^3 + \ln(a^2 - 3a + 3), & 0 \leq x < 3 \\ x - 18, & x \geq 3 \end{cases}$ complete sets of values of 'a' such that $f(x)$ has a local minima at $x = 3$ is	B
	$[-1, 2]$	
	$(-\infty, 1] \cup [2, \infty)$	
	$[1, 2]$	
	$(-\infty, 1) \cup (2, \infty)$	
Sol.	Clearly, $f(x)$ is decreasing just before $x = 3$ and increasing after $x = 3$. For $x = 3$ to be the point of local minima, $f(3) < f(3^-)$. or $-15 < 12 - 27 + \ln(a^2 - 3a + 3)$ or $a^2 - 3a + 3 > 1$ or $a \in (-\infty, 1) \cup (2, \infty)$	

52.	Let the foci of the ellipse $\frac{x^2}{16} + \frac{y^2}{7} = 1$ and the hyperbola $\frac{x^2}{144} - \frac{y^2}{\alpha} = \frac{1}{25}$ coincide. Then the length of the latus rectum of the hyperbola is:	D
	$\frac{32}{9}$	
	$\frac{18}{5}$	
	$\frac{27}{4}$	
	$\frac{27}{10}$	
Sol.	Eccentricity of ellipse = $\frac{3}{4}$, Foci = $(\pm 3, 0)$ Eccentricity and Foci of hyperbola be $\frac{1}{12}\sqrt{144 + \alpha}$ and $\left(\pm \frac{12\sqrt{144 + \alpha}}{12}, 0\right)$ Since foci coincide $\therefore 3 = \frac{1}{5}\sqrt{144 + \alpha} \Rightarrow \alpha = 81$, Length of latus rectum = $\frac{2b^2}{a} = \frac{27}{10}$	
53.	Vectors $\vec{OA}, \vec{OB}, \vec{OC}$ are on the same plane, $\vec{OA} = 1, \vec{OB} = 1, \vec{OC} = \sqrt{2}$, the relative position is shown in the figure.  If $\tan \alpha = 7$ and $(\vec{OB}, \vec{OC}) = \frac{\pi}{4}$ and $\vec{OC} = m\vec{OA} + n\vec{OB}$, ($m, n \in R$), find the value of $(m + n)$.	D
	7	
	9	
	8	
	3	
Sol.	$\vec{OC} = m\vec{OA} + n\vec{OB}$ $\vec{c} = m\vec{a} + n\vec{b}$ Given $\vec{a} = 1, \vec{b} = 1, \vec{c} = \sqrt{2}, \tan \alpha = 7$ Now take dot of equation (1) with $\vec{a}$ and $\vec{b}$ to get $m = 5/4; n = 7/4$ $\therefore m + n = 3$	
54.	Consider the function $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)}, & x \neq 2 \\ 7, & x = 2 \end{cases}$, where $P(x)$ is a polynomial such that $P''(x)$ is always a constant and $P(3) = 9$. If $f(x)$ is continuous at $x = 2$, then $P(4)$ is equal to	B
	20	
	22	
	24	
	25	
Sol.	$P''(x)$ is constant $\Rightarrow P(x)$ is quadratic	
55.	In a paper of English there are 5 questions such that the sum of marks is 30 and the marks for any question is not less than 2 and not more than 8. If the number of ways in which marks can be awarded is a 3 digit number xyz then the value	B

	of $\frac{2}{5}(x + y + z)$ is equal to? (Given that marks can be allotted in integers only)	
	5.4	
	6.4	
	7.4	
	8.4	
Sol.	Required no. of ways = coeff of $x^{30} \text{ in } (x^2 + x^3 + \dots + x^8)^5 = \text{coeff of } x^{30} \text{ in } \left(\frac{x^2(1-x^7)}{1-x} \right)^5$ $= \text{coeff of } x^{20} \text{ in } (1-x^7)^5 (1-x)^{-5} = {}^{24}C_{20} - 5 {}^{17}C_{13} + 10 {}^{10}C_6 = 826$	
56.	Let a, b, c (in order) be the first three terms of a sequence and satisfying $\log\left(\frac{8b^3 - a^3 - c^3}{6abc}\right) = 0$ and $\log b = \log(a^2 - 4) = \log(c - 2)$. If T_n and S_n denote n^{th} term and sum of first n terms of the sequence, then wrong statement will be	A
	$T_{10} = 31$	
	$S_{10} = 120$	
	$T_{21}^2 - 2T_{21}T_1 + 2T_1^2 = 1609$	
	$S_{11} - S_{10} = 23$	
Sol.	$8b^3 - a^3 - c^3 = 6abc$ $a^3 + c^3 + (-2b)^3 = 3ab(-2b)$ $\Rightarrow a + c - 2b = 0$ (or) $a = c = -2b$	
57.	If $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$ is a relation on the set of integers $\mathbb{Z}$, then the domain of R^{-1} is	D
	$\{-2, -1, 1, 2\}$	
	$\{0, 1\}$	
	$\{-2, -1, 0, 1, 2\}$	
	$\{-1, 0, 1\}$	
Sol.	$\{-1, 0, 1\}$ $R = \{(x, y) : x, y \in \mathbb{Z}, x^2 + 3y^2 \leq 8\}$  For domain of R^{-1} Collection of all integral of y's For $x = 0, 3y^2 \leq 8$ $\Rightarrow y \in \{-1, 0, 1\}$	
58.	If $\sum_{i=1}^9 (x_i - 5) = 9$ and $\sum_{i=1}^9 (x_i - 5)^2 = 45$ then the standard deviation of	C

	the 9 items $x_1, x_2, \dots, x_9$ is	
	9	
	4	
	2	
	3	
Sol.	We have, $\sum_{i=1}^9 (x_i - 5) = 9$ and $\sum_{i=1}^9 (x_i - 5)^2 = 45$ $SD = \sqrt{\frac{\sum_{i=1}^9 (x_i - 5)^2}{9} - \left(\frac{\sum_{i=1}^9 (x_i - 5)}{9} \right)^2}$ $\Rightarrow SD = \sqrt{\frac{45}{9} - \left(\frac{9}{9} \right)^2}$ $\Rightarrow SD = \sqrt{5 - 1} = \sqrt{4} = 2$	
59.	Given set = $\{1, 2, 3, 4, 5, \dots, 1000\}$. One number is selected randomly from the set. Then the probability that selected number is divisible by at least one 2, 3 or 5.	A
	<u>367</u>	
	<u>500</u>	
	<u>257</u>	
	<u>500</u>	
	<u>417</u>	
	<u>500</u>	
	<u>217</u>	
	<u>500</u>	
Sol.	Let A_k be the of positive integers from 1 to 1000, which are divisible by k. If $[.]$ denotes the greatest integer function, then $n(A_2) = \left[\frac{1000}{2} \right] = 500$ $n(A_3) = \left[\frac{1000}{3} \right] = 333$ $n(A_5) = \left[\frac{1000}{5} \right] = 200$ Also $n(A_2 \cap A_3) = \left[\frac{1000}{6} \right] = 166$ $n(A_3 \cap A_5) = \left[\frac{1000}{15} \right] = 66$ $n(A_2 \cap A_5) = \left[\frac{1000}{10} \right] = 100$ and $n(A_2 \cap A_3 \cap A_5) = \left[\frac{1000}{30} \right] = 33$ So, $n(A_2 \cup A_3 \cup A_5) = 500 + 333 + 200 - 166 - 66 - 100 + 33 = 734$	
60.	Let $P(\alpha, \beta, \gamma)$ be the image of the point $Q(3, -3, A)$ in the line $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1}$ and R be the point $(2, 5, -A)$. If the area of the triangle PQR is λ and $\lambda^2 = 14K$, then K is equal to:	A
	81	
	36	
	18	
	72	

Sol.	Let $\frac{x-0}{1} = \frac{y-3}{1} = \frac{z-1}{-1} = \lambda$ $x = \lambda, y = 3 + \lambda, z = 1 - \lambda$ $S = (\lambda, \lambda + 3, -\lambda + 1)$ d.r's of $SQ = (\lambda - 3, \lambda + 6, -\lambda)$ d.r's of line $= (1, 1, -1)$ $\overline{SQ} \perp \text{line} \Rightarrow 1(\lambda - 3) + 1(\lambda + 6) - 1(-\lambda) = 0$ $\Rightarrow 3\lambda + 3 = 0 \Rightarrow \lambda = -1$ $S = (-1, 2, 2)$ $(-1, 2, 2) = \text{mid point of } PQ = \left(\frac{\alpha + 3}{2}, \frac{\beta - 3}{2}, \frac{\gamma + 1}{2} \right)$ $-1 = \frac{\alpha + 3}{2} \quad 2 = \frac{\beta - 3}{2} \quad 2 = \frac{\gamma + 1}{2}$ $\alpha = -5 \quad \beta = 7 \quad \gamma = 3$ $\therefore P = (-5, 7, 3)$ Area of $\Delta PQR = \frac{1}{2} \overline{PR} \times \overline{PQ}$ $\overline{PQ} = (+8, -10, -2)$ $\overline{PR} = (7, -2, -4)$ area of $\Delta PQR = \frac{1}{2} \overline{PR} \times \overline{PQ}$ $\lambda = \frac{1}{2} \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ 8 & -10 & -2 \\ 7 & -2 & -4 \end{vmatrix}$ $= \frac{1}{2} [\vec{i}(40 - 4) - \vec{j}(-32 + 14) + \vec{k}(-16 + 70)]$ $= \frac{1}{2} [36\vec{i} + 18\vec{j} + 54\vec{k}]$ $\lambda = \frac{1}{2} \sqrt{1296 + 324 + 2916} = \frac{1}{2} \sqrt{4536}$ $\lambda^2 = \frac{4536}{4} = 1134$ $14K = \frac{4536}{4} = 1134$ $K = \frac{4536}{4} = 81$	
61.	Suppose the solution of the differential equation $\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ represents a circle passing through the origin. Then the radius of this circle is:	D
	$\sqrt{17}$	
	2	
	$\frac{1}{2}$	
	$\frac{\sqrt{17}}{2}$	

Sol.	$\frac{dy}{dx} = \frac{(2+\alpha)x - \beta y + 2}{\beta x - 2\alpha y - (\beta\gamma - 4\alpha)}$ $\Rightarrow \beta x dy - 2\alpha y dy - (\beta\gamma - 4\alpha)dy = (2+\alpha)x dx - \beta y dx + 2 dx$ $\Rightarrow (2+2)x dx - \beta(x dy + y dx) + 2 dx + 2\alpha y dy + (\beta\gamma - 4\alpha)dy = 0$ Integrating on both sides $\Rightarrow (2+\alpha) \int x dx - \beta \int d(xy) + 2 \int 1 dx + 2\alpha \int y dy + (\beta\gamma - 4\alpha) \int 1 dy = 0$ $\Rightarrow (2+\alpha) \frac{x^2}{2} - \beta xy + 2x + 2\alpha \frac{y^2}{2} + (\beta\gamma - 4\alpha)y = c \text{ ---- (1)}$ This equation represent a circle coefficient of x^2 = coefficient of $y^2 \Rightarrow \frac{2+\alpha}{2} = \alpha \Rightarrow 2+\alpha = 2\alpha$ $\alpha = 2$ coefficient of $xy = 0 \Rightarrow \beta = 0$ $(1) \Rightarrow 2x^2 + 2y^2 + 2x - 8y = c$ since circle passes (0,0) $c = 0$ $\therefore \text{ circle } 2x^2 + 2y^2 + 2x - 8y = 0$ $\Rightarrow x^2 + y^2 + x - 4y = 0$ $c = \left(-\frac{1}{2}, 2\right)$ $r = \sqrt{\frac{1}{4} + 4} = \sqrt{\frac{17}{4}} = \frac{\sqrt{17}}{2}$	
62.	Let α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$. If $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ is equal to:	B
	1	
	4	
	6	
	7	
Sol.	α, β be roots of $x^2 + \sqrt{2}x - 8 = 0$ $U_n = \alpha^n + \beta^n$, then $\frac{U_{10} + \sqrt{2}U_9}{2U_8}$ α is a root of $x^2 + \sqrt{2}x - 8 = 0$ $\alpha^2 + \sqrt{2}\alpha = 8$ $\alpha^8(\alpha^2 + \sqrt{2}\alpha) = \alpha^8 \cdot 8$ $\alpha^{10} + \sqrt{2}\alpha^9 = 8\alpha^8$ Similarly $\beta^{10} + \sqrt{2}\beta^9 = 8\beta^8$ Add $\alpha^{10} + \beta^{10} + \sqrt{2}(\alpha^9 + \beta^9) = 8(\alpha^8 + \beta^8)$ $U_{10} + \sqrt{2}U_9 = 8U_8$ $\frac{U_{10} + \sqrt{2}U_9}{2U_8} = \frac{8}{2} = 4$	
63.	If z_1, z_2 are two distinct complex numbers such that $\left \frac{z_1 - 2z_2}{\frac{1}{z_1} - z_2} \right = 2$, then	A
	Either z_1 lies on a circle of radius 1 or z_2 lies on a circle of radius $\frac{1}{2}$	
	z_1 lies on a circle of radius $\frac{1}{2}$ and z_2 lies on a circle of radius 1	
	Either z_1 lies on a circle of radius $\frac{1}{2}$ or z_2 lies on a circle of radius 1	
	Both z_1 and z_2 lie on the same circle	

Sol.	$ z_1 - 2z_2 = 1 - 2\bar{z}_1 z_2 $ S.O.B.S $ z_1 - 2z_2 ^2 = 1 - 2\bar{z}_1 z_2 ^2$ $z ^2 = z \cdot \bar{z}$ $(z_1 - 2z_2)(\bar{z}_1 - 2\bar{z}_2) = (1 - 2\bar{z}_1 z_2)(1 - 2\bar{z}_1 z_2)$ $\Rightarrow z_1 ^2 + 4 z_2 ^2 - 2z_1 \bar{z}_2 - 2\bar{z}_1 z_2$ $= 1 - 2z_1 \bar{z}_2 - 2\bar{z}_1 z_2 + 4 z_1 ^2 z_2 ^2$ $\Rightarrow z_1 ^2 + 4 z_2 ^2 - 4 z_1 ^2 z_2 ^2 - 1 = 0$ $\Rightarrow (z_1 ^2 - 1)(4 z_2 ^2 - 1) = 0$ $\Rightarrow z_1 = 1, z_2 = \frac{1}{2}$																			
64.	If all the words with or without meaning made using all the letters of the word “NAGPUR: are arranged as in a dictionary, then the word at 315 th position in this arrangement is:	D																		
	NRAPUG																			
	NRAGUP																			
	NRAGPU																			
	NRAPGU																			
Sol.	<table><tr><td>5</td><td>4</td><td>3</td><td>2</td><td>1</td><td>0</td></tr><tr><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td><td>X</td></tr><tr><td>2</td><td>0</td><td>0</td><td>0</td><td>0</td><td>1</td></tr></table> $5 \times 2 + 0 \times 1 = 240 + 1 = 241$ $315 - 241 = 74$ $74 = 4 \times 3 + 2 \times 1$ $\therefore$ NRAPGU is correct answer	5	4	3	2	1	0	X	X	X	X	X	X	2	0	0	0	0	1	
5	4	3	2	1	0															
X	X	X	X	X	X															
2	0	0	0	0	1															
65.	Let [t] denote the largest integer less than or equal to t. If $\int_0^3 \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx = a + b\sqrt{2} - \sqrt{3} - \sqrt{5} + c\sqrt{6} - \sqrt{7}$, where $a, b, c \in \mathbb{Z}$ then $a + b + c$ is equal to _____.	A																		
	23																			
	21																			
	25																			
	26																			
Sol.	$\int_0^1 \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_1^{\sqrt{2}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_{\sqrt{2}}^{\sqrt{3}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx$ $+ \int_{\sqrt{3}}^{\sqrt{4}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_{\sqrt{4}}^{\sqrt{5}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_{\sqrt{5}}^{\sqrt{6}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx$ $+ \int_{\sqrt{6}}^{\sqrt{7}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_{\sqrt{7}}^{\sqrt{8}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx + \int_{\sqrt{8}}^{\sqrt{9}} \left([x^2] + \left\lceil \frac{x^2}{2} \right\rceil \right) dx$ $= \int_0^1 0 dx + \int_1^{\sqrt{2}} 1 dx + \int_{\sqrt{2}}^{\sqrt{3}} 3 dx + \int_{\sqrt{3}}^{\sqrt{4}} 4 dx + \int_{\sqrt{4}}^{\sqrt{5}} 6 dx + \int_{\sqrt{5}}^{\sqrt{6}} 7 dx + \int_{\sqrt{6}}^{\sqrt{7}} 9 dx + \int_{\sqrt{7}}^{\sqrt{8}} 10 dx + \int_{\sqrt{8}}^{\sqrt{9}} 12 dx$ $= 31 - 6\sqrt{2} - \sqrt{3} - \sqrt{5} - \sqrt{7} - 2\sqrt{6}$ $a = 31, b = -6, c = -2$ $a + b + c = 31 - 6 - 2 = 23$																			
66.	If A is a square matrix of order 3 such that $\det(A) = 3$ and $\det \left(\text{adj} \left(-4 \text{adj} \left(-3 \text{adj} \left(3 \text{adj} (2A)^{-1} \right) \right) \right) \right) = 2^m 3^n$, then $m + 2n$ is equal to:	B																		

	2	
	4	
	6	
	3	
Sol.	$\left \left(adj \left(-4adj \left(-3adj \left(3adj(2A)^{-1} \right) \right) \right) \right) \right ^2$ $\left -4adj \left(-3adj \left(3adj(2A)^{-1} \right) \right) \right ^2$ $4^6 \left adj \left(-3adj \left(3adj(2A)^{-1} \right) \right) \right ^2$ $4^6 \left(\left -3adj \left(3adj(2A)^{-1} \right) \right ^2 \right)^2$ $4^6 \left -3adj \left(3adj(2A)^{-1} \right) \right ^4$ $4^6 \times 3^{12} \left adj \left(3adj(2A)^{-1} \right) \right ^4$ $4^6 \times 3^{12} \left 3adj(2A)^{-1} \right ^8$ $4^6 \times 3^{12} \times 3^{24} \left adj(2A)^{-1} \right ^8$ $4^6 \times 3^{36} \left (2A)^{-1} \right ^{16}$ $4^6 \times 3^{36} \times \frac{1}{2^{48} A ^{16}}$ $2^{12} \times 3^{36} \times \frac{1}{2^{48} \times 3^{16}}$ $2^{-36} \times 3^{20} = 2^m \times 3^n$ $m = -36, n = 20$ $m + 2n = -36 + 2(20)$ $= -36 + 40$ $= 4$	
67.	The coefficient of x^{70} in $x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + \dots + x^{54}(1+x)^{46}$ is ${}^{99}C_p - {}^{46}C_q$ then a possible value of $p + q$ is	D
	68	
	61	
	55	
	83	
Sol.	$x^2(1+x)^{98} + x^3(1+x)^{97} + x^4(1+x)^{96} + \dots - x^{54}(1+x)^{46}$ $\begin{array}{ccccccc} \downarrow & & \downarrow & & \downarrow & & \downarrow \\ x^{68} & & x^{67} & & x^{66} & & x^{16} \end{array}$ Coefficient $x^{70} = 98C_{68} + 97C_{67} + 96C_{66} + \dots - 47C_{17} + 46C_{16}$ $= 46C_{15} + 46C_{16} + 47C_{17} + \dots - 98C_{68} - 46C_{15}$ $= 99C_{68} - 46C_{45} (\because nC_r + nC_{r-1} = n+1C_r)$ $P = 68, q = 15$ $p = q = 83$	

68.	If the sum of the series $\frac{1}{1.(1+d)} + \frac{1}{(1+d)(1+2d)} + \dots + \frac{1}{(1+9d)(1+10d)}$ is equal to 5, then 50 d is equal to	C
	10	
	20	
	5	
	15	
Sol.	$\frac{1}{d} \left[\frac{d}{1(1+d)} + \frac{d}{(1+d)(1+2d)} + \dots + \frac{d}{(1+9d)(1+10d)} \right] = 5$ $\frac{1}{d} \left[\frac{(1+d)-1}{1(1+d)} + \frac{(1+2d)-(1+d)}{(1+d)(1+2d)} + \dots + \frac{(1+10d)-(1+9d)}{(1+9d)(1+10d)} \right] = 5$ $\frac{1}{d} \left[\frac{1}{1} - \frac{1}{1+d} - \frac{1}{1+2d} + \dots + \frac{1}{1+9d} - \frac{1}{1+10d} \right] = 5$ $\frac{1}{d} \left[\frac{1}{1} - \frac{1}{1+10d} \right] = 5$ $\frac{1}{d} \left[\frac{10d}{1+10d} \right] = 5 \Rightarrow 2 = 1+10d$ $10d = 1$ $d = \frac{1}{10}$ $50d = 5$	
69.	If the domain of the function $f(x) = \sin^{-1} \left(\frac{x-1}{2x+3} \right)$ is $R - (\alpha, \beta)$ then $12\alpha\beta$ is equal to	D
	24	
	36	
	40	
	32	
Sol.	$f(x) = \sin^{-1} \left(\frac{x-1}{2x+3} \right)$ $-1 \leq \frac{x-1}{2x+3} \leq 1$ $-2x-3 \leq x-1$ $-3x-2 \leq 0$ $-3x \leq 2$ $x \leq \frac{-2}{3}$ $x-1 \leq 2x+3$ $0 \leq x+4$ $-4 \leq x$ $\text{Domain} = (-\infty, -4] \cup \left[\frac{-2}{3}, \infty \right)$ $\Rightarrow R - \left(-4, \frac{-2}{3} \right)$ $\alpha = -4, \beta = \frac{-2}{3}$ $12\alpha\beta = 12 \times -4 \times \frac{2}{3}$ $12\alpha\beta = 32$	
70.	Let $\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2} (\alpha x + \log_e \beta \sin x + \gamma \cos x) + C$, where C is the constant of integration then $\alpha + \frac{\gamma}{\beta}$ is equal to	B
	3	
	4	

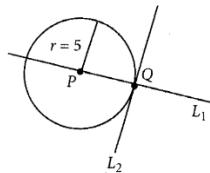
	7	
	1	
Sol.	$\int \frac{2 - \tan x}{3 + \tan x} dx = \frac{1}{2}(\alpha x + \log_e \beta \sin x + \gamma \cos x) + c$ $\frac{2 - \sin x}{3 + \frac{\sin x}{\cos x}} = \int \frac{2 \cos x - \sin x}{3 \cos x + \sin x}$ $2 \cos x - \sin x = A(3 \cos x + \sin x) + B(\cos x - 3 \sin x)$ By comparing Coefficient $3A + 2B = 2 \rightarrow (1)$ $A - 3B = -1 \rightarrow (2)$ Solving (1) & (2) $a - 3b = -1$ $9a + 3b = 6$ $10a = 5$ $A = \frac{1}{2}$ $B = \frac{1}{2}$ $a = 1, \beta = 1, \gamma = 3$ $\alpha + \frac{8}{\beta} = 1 + 3 = 4$	
71.	If the area of the region $\{(x, y) \in R^2 : y^2 \geq 4x, y \leq \frac{x}{2} + 2\}$ is Δ , then the value of $[\Delta]$. [Note: $[x]$ denotes greatest integer less than or equal to x .]	10
Sol.	Area of the shaded region is $= \frac{1}{2} \times 8 \times 8 - \frac{2}{3} \times 32$ $= \frac{1}{3} \times 32$ $\therefore \frac{32}{3} = \Delta$ $\therefore [\Delta] = 10$	
72.	Let (x, y) satisfy the curve $x^2 + y^2 - 6x - 8y + 21 = 0$. Let m_1 and m_2 are minimum and maximum values of $\left(\frac{x}{y}\right)$. If $m_1^2 + m_2^2 = \frac{p}{q}$ where $p, q \in N$, then the least value $(p + q)$ is.	25
Sol.	$\frac{x}{y} = m$ Let $x = my$ be the tangent Applying $\frac{ 3 - 4m }{\sqrt{1 + m^2}} = 2$ $12m^2 - 24m + 5 = 0$ $(m_1 + m_2)^2 - 2m_1m_2 = \left(\frac{24}{12}\right)^2 - 2 \times \frac{5}{12}$ $= \frac{19}{6}$	
73.	If the system of equations $2x + 7y + \lambda z = 3$	38

	$3x + 2y + 5z = 4$ $x + \mu y + 32z = -1$ has infinitely many solutions, then $(\lambda - \mu)$ is equal to _____.	
Sol.	$2x + 7y + \lambda z = 3$ $3x + 2y + 5z = 4$ $x + \mu y + 32z = -1$ $\Delta = 0$ $\begin{vmatrix} 2 & 7 & -1 \\ 3 & 2 & 5 \\ 1 & \mu & 32 \end{vmatrix} = 0$ $2(64 - 5\mu) - 7(96 - 5) - 1(3\mu - 2) = 0$ $128 - 10\mu - 637 - 3\mu + 2 = 0$ $-507 = 13\mu \Rightarrow \mu = -39$ $\Delta_2 = \begin{vmatrix} 2 & 3 & \lambda \\ 3 & 4 & 5 \\ 1 & -1 & 32 \end{vmatrix} = 0$ $2(128 + 5) - 3(96 - 5) + \lambda[-3 - 4] = 0$ $266 - 273 - 7\lambda = 0$ $-7 - 7\lambda = 0$ $-7\lambda = 7 \Rightarrow \lambda = -1$ $\lambda - \mu = -1 - (-39) = 38$	
74.	If the vectors, $\vec{p} = (a + 1)\hat{i} + a\hat{j} + a\hat{k}$, $\vec{q} = a\hat{i} + (a + 1)\hat{j} + a\hat{k}$ and $\vec{r} = a\hat{i} + a\hat{j} + (a + 1)\hat{k}$, ($a \in R$) are coplanar and $3(\vec{p} \cdot \vec{q})^2 - \lambda \vec{r} \times \vec{q} ^2 = 0$, then the value of λ is _____.	1
Sol.	(1) : Since, $\vec{p}, \vec{q}$ and $\vec{r}$ are coplanar. $\therefore \begin{vmatrix} a+1 & a & a \\ a & a+1 & a \\ a & a & a+1 \end{vmatrix} = 0$ Applying $R_2 \rightarrow R_2 - R_1, R_3 \rightarrow R_3 - R_1$, $\begin{vmatrix} a+1 & a & a \\ -1 & 1 & 0 \\ -1 & 0 & 1 \end{vmatrix} = 0$ $\Rightarrow a+1+a+a=0 \Rightarrow a=-\frac{1}{3}$ $\therefore \vec{p} = \frac{1}{3}(2\hat{i} - \hat{j} - \hat{k}); \vec{q} = \frac{1}{3}(-\hat{i} + 2\hat{j} - \hat{k})$ and $\vec{r} = \frac{1}{3}(-\hat{i} - \hat{j} + 2\hat{k})$ Now, $\vec{p} \cdot \vec{q} = \frac{1}{9}(-2 - 2 + 1) = -\frac{1}{3}$ and $\vec{r} \times \vec{q} = \frac{1}{9} \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & -1 & 2 \\ -1 & 2 & -1 \end{vmatrix}$ $= \frac{1}{9} [\hat{i}(1-4) - \hat{j}(1-2) + \hat{k}(-2-1)]$ $= \frac{1}{9} (-3\hat{i} - 3\hat{j} - 3\hat{k}) = -\frac{\hat{i} + \hat{j} + \hat{k}}{3}$ Now, $ \vec{r} \times \vec{q} = \frac{1}{3}\sqrt{3} \Rightarrow \vec{r} \times \vec{q} ^2 = \frac{1}{3}$ Given, $3(\vec{p} \cdot \vec{q})^2 - \lambda \vec{r} \times \vec{q} ^2 = 0$ $\Rightarrow 3 \cdot \frac{1}{9} - \lambda \cdot \frac{1}{3} = 0 \Rightarrow \lambda = 1$	
75.	Let a circle C of radius 5 lie below the x-axis. The line $L_1: 4x + 3y + 2 = 0$ passes through the centre P of the circle C and intersects the line $L_2: 3x - 4y - 11 = 0$ at Q . The line L_2 touches C at the point Q . Then the distance of P from the line $5x - 12y + 51 = 0$ is _____.	11

Sol.

(11) : Let $r = 5$, $L_1 : 4x + 3y + 2 = 0 \dots(i)$

$L_2 : 3x - 4y - 11 = 0 \dots(ii)$



Solving equation (i) and (ii), we get

$$x = 1, y = -2$$

$$\therefore Q \equiv (1, -2)$$

Let the coordinate of P be $\left(\lambda, \frac{-1}{3}(2 + 4\lambda)\right)$

Now, $PQ = 5$

$$(\lambda - 1)^2 + \left(\frac{-1}{3}(2 + 4\lambda) + 2\right)^2 = 25$$

$$\lambda^2 + 1 - 2\lambda + \frac{1}{9}(2 + 4\lambda)^2 + 4 - \frac{4(2 + 4\lambda)}{3} = 25$$

$$\Rightarrow \lambda = 4, -2.$$

Since the circle C lie below the x-axis.

$$\therefore \lambda = 4$$

Thus, $P \equiv (4, -6)$

The distance of $P(4, -6)$ from the line

$5x - 12y + 51 = 0$ is

$$d = \frac{20 + 72 + 51}{\sqrt{25 + 144}} = 11$$